

Risk Based Capital Allocation Framework for Morpho Vault Curation

Feather

Overview:

This document outlines a quantitative methodology for the risk-aware curation and capital distribution of isolated lending markets within Morpho vaults. The framework is built to uphold four core constraints: liquidation exposure tolerance, market rate competitiveness, optimal utilization, and withdrawal liquidity availability. Asset-specific risk scores drive allocation decisions, integrating volatility, liquidity, oracle reliability, utilization behavior, and simulation-derived stress test outcomes.

Composite Risk Score and Capital Ceiling:

In order to begin defining the relative risk for each market to which supply can be allocated, a Composite Risk Score (CRS), R_i , is calculated for each market i , derived from, but not limited to, the following normalized inputs and their associated tunable weights. This methodology draws conceptual influence from Mars' Protocol Risk Framework [12], though it differs significantly in logic and implementation

- α * Price Risk Metric Score:
 - $CVaR_{99\%, 90d}$: Tail risk from extreme price drawdowns [1]
 - $RollingReturnZScore_{7d}$: Short-term mean return reversion
 - $MaxIntradayDrawdown_{90d}$: Flash crash exposure
- β * Liquidity Risk Metric Score:
 - $Amihud_{90d}$: Illiquidity cost per volume traded [2]
 - $Depth_{2\%}$: Order book buffer at 2% price impact
 - $MaxLiquidityDrop_{99\%, 90d}$: Fragility of DEX depth under stress
- δ * Oracle Risk Metric Score:
 - $OracleDeviation_{30d}$: Oracle reliability risk
- ϵ * Borrow Instability Risk Metric Score:
 - $BorrowConcentrationIndex$: Whale liquidation vulnerability (inequality in loan distribution) [3]
 - $UtilizationVolatility_{30d}$: Volatility in utilization rate
 - $BorrowRateVolatility_{30d}$: Borrow demand instability

All variables are normalized on a 0-1 scale using min-max normalization. Min max normalization dynamically reflects real-world market conditions which enables more responsive allocation decisions. In order to minimize unstable sensitivity during normalization of metric values, a percentile-based bounds is used to determine min and max values used for normalization. This ensures outliers don't permanently skew normalized values while still incorporating meaningful shifts in markets over time. Normalization bounds are recalibrated on a weekly basis.

Normalized scores are then summed using the above weightings to produce R_i .

$$R_i = \alpha * P_i + \beta * L_i + \delta * O_i + \varepsilon * B_i$$

$$\alpha + \beta + \delta + \varepsilon = 1$$

Markets are tiered as follows:

- Tier 1: $R_i < 0.25$ – Highly allocatable
- Tier 2: $0.25 < R_i < 0.50$ – Moderately allocatable
- Tier 3: $0.50 < R_i < 0.75$ – Minorly allocatable
- Tier 4: $0.75 < R_i$ – Excluded from allocation

Once an R_i has been calculated for each market, a relative allocation weight is computed using inverse risk scores:

$$Weight_i = \frac{\frac{1}{R_i}}{\sum_j \frac{1}{R_j}}$$

Each market is assigned a capital ceiling, which serves as the maximum amount of capital that can be allocated to each isolated market based on core price and liquidity risk factors. This allocation ceiling prevents overexposure to high-risk, low-liquidity markets and is determined as follows:

$$CapitalCeiling_i = Weight_i * VaultTVL$$

R_i should be updated on an hourly basis based on a rolling 30-90 day window in order to ensure dynamic rescoring and reallocation based on evolving market conditions.

Tuning CRS weights:

The weights associated with various risk metrics — denoted as α (price risk), β (liquidity risk), δ (oracle risk), and ε (borrowing risk) — play a critical role in how the CRS reflects underlying market risk. A well-calibrated CRS ensures:

- Accurate *CapitalCeiling* calculations
- Correct simulation triggers
- More reliable capital allocation decisions

We perform empirical weight tuning using historical data so that the CRS maximally distinguishes risky markets from healthy ones. The historical dataset is composed of historical market states with known outcomes (e.g., instances of insolvency, liquidation spikes, or capital impairment). For each candidate set of weights, the CRS is calculated for all historical observations.

In order to evaluate the ability of the CRS to rank and separate risky markets from stable ones, we define a multi-objective scoring function, $WeightScore_{CRS_e}$ with the following normalized scoring metrics:

Objective	Variable	Description
AUC-ROC [4]	AUC_ROC	Measures global separability between risky and safe markets; higher is better.
Precision at Top-K [5]	Precision@TopK	Measures proportion of true risky markets in top-k CRS ranked markets
Mean Separation [6]	MeanSep	Measures the average CRS difference between risky and safe labeled classes.

$$WeightScore_{CRS} = \lambda_1(AUC_ROC) + \lambda_2(Precision@TopK) + \lambda_3(MeanSep)$$

$$\lambda_1 + \lambda_2 + \lambda_3 = 1$$

The CRS scoring function with predetermined scoring weights is then optimized by using an open-source hyperparameter optimization software such as Optuna [7]. Once a larger labeled dataset of historical outcomes is created, the CRS scoring weights can be refined.

The use of an optimization software like Optuna allows for the efficient setup and interactive exploration of the space of CRS scoring weight configurations and builds probabilistic models of scoring weight combinations that yield strong separation between risky and safe vaults, distinguishing them from less predictive combinations.

This results in a scoring configuration that:

- Maximizes the model's ability to separate high-risk from low-risk markets
- Improves the reliability of CRS as indicator of adverse market performance

Simulation Engine Methodology:

The goal of this simulation framework is to model the behavior of lending vaults under adverse market conditions in order to inform risk-adjusted capital rebalancing. A Monte Carlo simulation [8] is a computational technique used to model outcome probability in uncertain scenarios via random

sampling, and has become commonly used to manage risk within portfolios of assets . This technique is applied within this framework to assess the probability distribution of risky outcomes under thousands of potential future scenarios involving price and liquidity shocks.

Simulations are triggered based on the following events:

- R_i based: When time based rebalances occur as a result of an improved R_i
- Time based: every 12 hrs

For the Monte Carlo Simulation Engine, N random price paths are generated for each collateral and debt asset using one of the most fundamental models for valuing assets, Geometric Brownian Motion (GBM) [9].

In order to simulate a wide variety of random liquidity shock scenarios, N liquidity decay paths are sampled from a Beta (α, β) distribution for each collateral and debt asset. The Beta Distribution is a continuous probability distribution of a standard uniform random sample on the interval [0, 1] making it useful for the modeling of percentages or fractional values such as random percent reductions in liquidity [10]. Its shape is governed by the parameters α and β , allowing flexible modeling of different market risk profiles.

After calculating the stressed depth of liquidity and price for a path, the LTVs for each borrower in that particular market are recalculated. If a recalculated LTV is greater than the liquidation threshold for that market, a liquidation is simulated. These simulation events are aggregated over N number of paths for all borrow positions to determine the total liquidation volume used to calculate potential collateral shortfalls.

For N number of paths, the simulations will yield:

- Insolvency Rate - used as trigger for rebalancing when threshold crossed
- Collateral Shortfall_{95%} - used for scaling factor during allocation upon rebalancing

If Insolvency rate is higher than its associated threshold, the CapitalCeiling will be scaled down as a function of the collateral shortfall relative to its associated threshold (expressed as a percentage). Otherwise, capital may be up to the CapitalCeiling if the R_i has improved and no adverse outcomes are detected during simulations.

$$ScaledCapitalCeiling (SCC) = \min(CapitalCeiling * \frac{CollateralShortfall_{95\%}}{CollateralShortfall_{95\%} threshold}, CapitalCeiling)$$

Operational Constraint Optimization:

Prior to rebalancing capital allocations within a vault, ScaledCapitalCeilings are optimized using multi-objective optimization to satisfy operational constraints relating to maintaining competitive borrow and utilization rates as well as idle liquidity buffers.

This methodology for vault curation and capital distribution enforces the following operational constraints inspired by prior work from Gauntlet [11]:

- Rate Competitiveness: Resulting supply and borrow rates must remain within a target deviation from comparable on-chain lending markets.
- Utilization Band: Utilization of each market must remain within 85-92% to optimize yield and availability.
- Withdrawal Liquidity: At least 10% of total vault deposits must be available for immediate withdrawal at any given time.

We define a multi-objective scoring function $OperationalScore(a^{\rightarrow})$ that evaluates each potential allocation vector $a^{\rightarrow}$ using a weighted combination of operational objectives. The allocation vector that maximizes this score, subject to capital ceilings, is selected.

Objective	Variable	Description
Maximize Allocated Capital	A_i	capital deployment efficiency
Target Utilization	$U_i(A_i)$	closeness to target utilization range
Target Borrow Rate	$R_i(A_i)$	expected borrow rate competitiveness
Maintain Idle Liquidity Buffer	$B_i(A_i) = D_i - A_i$	adequacy of idle liquidity buffers

$$OperationalScore(a^{\rightarrow}) = w_1 S_{alloc}(a^{\rightarrow}) + w_2 S_{util}(a^{\rightarrow}) + w_3 S_{rate}(a^{\rightarrow}) + w_4 S_{buffer}(a^{\rightarrow})$$

$$w_1 + w_2 + w_3 + w_4 = 1$$

This OperationalScore function is then optimized by tuning the associated operational score weights using the same open-source hyperparameter optimization software for optimizing the CRS scoring function, Optuna. This allows for the maximization of the operational scoring function (higher adherence to operational constraints) across valid allocation vectors $a^{\rightarrow}$ within vault i , under the constraint:

$$\sum_i a_i \leq Total\ Capital\ Available\ for\ vault\ i, \quad 0 \leq a_i \leq SCC_i$$

This results in an allocation vector that:

- Honors all risk-based ScaledCapitalCeilings from simulation analysis
- Achieves the best adherence to operational targets

Rebalancing Methodology:

Capital is reallocated across vaults based on dynamic rebalancing logic informed by updated Composite Risk Scores (CRS), Monte Carlo Simulation (MCS) outputs, and adherence to operational constraints. Rebalancing ensures that capital exposure is aligned with evolving market conditions and the protocol's risk tolerance.

Rebalances in capital allocation are both event and schedule driven. The following trigger types cause rebalancing:

Trigger Type	Condition	Change (increase/decrease)
Time-based	Daily → Simulate allocation safety for assets with $+\Delta R_i$	both
ΔR_i -based	Negative ΔR_i exceeds threshold	decrease

All time-based rebalance triggers require simulation of adverse changes to liquidity and price scenarios to assess potential insolvency risk prior and optimization of operational constraints prior to capital allocation.

Equation Appendix:

Composite Risk Score (R_i):

$$R_i = \alpha * P_i + \beta * L_i + \delta * O_i + \varepsilon * B_i$$

Where:

- P_i = Average Price Risk Metric Score

$$P_i = \frac{(CVaR_{99\%, norm} + MaxDrawdown_{90d, norm} + RollingReturnZScore_{7d, norm})}{3}$$

- L_i = Average Liquidity Risk Metric Score

$$L_i = \frac{(MaxLiquidityDrop_{99\%, 90d, norm} + Depth_{2\%} + Amihud_{90d, norm})}{3}$$

- O_i = Average Oracle Risk Metric Score

$$O_i = OracleDeviation_{30d, norm}$$

- B_i = Average Borrowing Instability Risk Metric Score

$$B_i = \frac{(BorrowRateVolatility_{30d, norm} + BorrowConcentrationIndex_{norm} + UtilizationVolatility_{30d, norm})}{3}$$

- α = Price Risk Weight
- β = Liquidity Risk Weight
- δ = Oracle Risk Weight
- ε = Borrow Instability Risk Weight

Rolling Return Z-Score:

$$Z_i = \frac{r_i - \mu}{\sigma}$$

where:

- r_i = average return over short horizon (7d)
- μ = mean returns (90d)
- σ = standard deviation of returns (90d)

Conditional Value at Risk (CVaR), 99% confidence level:

$$\text{CVaR}_\alpha(L) = E [L \mid L \geq \text{VaR}_\alpha(L)]$$

$$\text{CVaR}_{99\%} = \frac{1}{k} * \sum_{i=1}^k r_i$$

Where:

- α = percentile of return distribution
- L = negative asset return
- $\text{VaR}_\alpha(L)$ = percentile value at risk of the return distribution
- $k = \lfloor n \cdot (1 - \alpha) \rfloor$
 - where n = total number of returns
- $r_1, \dots, r_k$ are the worst 1% of returns

Borrow Rate Volatility:

$$\sigma_r = \sqrt{\frac{1}{n-1} * \sum_{i=1}^n (r_i - \bar{r})^2}$$

Where:

- $\bar{r}$ is the average borrow rate (expressed as decimal) over n days
- r_i is the borrow rate on day i

Utilization Volatility:

$$\sigma_u = \sqrt{\frac{1}{n-1} * \sum_{i=1}^n (u_i - \bar{u})^2}$$

Where:

- $\bar{u}$ is the average utilization rate (expressed as decimal) over n days
- u_i is the utilization rate on day i

Borrow Concentration Index (Gini Coefficient for Borrow Concentration):

$$G = \frac{\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j|}{2n^2 * \bar{x}}$$

Where:

- x_i and x_j are individual borrow amounts for users i and j
- n is the total number of borrowers
- $\bar{x}$ is the mean borrow amount

Oracle Deviation:

$$d_t = \left| \frac{p_t^{oracle} - p_t^{spot}}{p_t^{spot}} \right|$$

- a) **Max Oracle Deviation** = $\max_t d_t$
b) **Average Oracle Deviation** = $\frac{1}{n} \sum_{t=1}^n d_t$

where:

- n = number of time points

Max Intraday Drawdown:

$$\text{Drawdown}_d = \frac{P_d^{low} - P_d^{high}}{P_d^{high}}$$

$$\text{MaxDrawdown}_{\text{intraday}} = \min_{d \in [1, n]} \text{Drawdown}_d$$

where:

- P_d^{low} is the lowest price of the asset on day d
- P_d^{high} is the highest price of the asset on day d
- n = number of days

Amihuds Illiquidity Measure:

$$I = \frac{1}{n} \sum_{i=1}^n \frac{|R_i|}{V_i}$$

Where:

- R_i is the daily asset return on day i
- V_i is the daily dollar trading volume on day i

Geometric Brownian Motion (GBM):

$$P_{t+1} = P_t * \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}Z_t\right), \quad Z_t \sim \mathcal{N}(0, 1)$$

Where:

- P_t = starting price of token at time t
- σ = historical volatility of token
- μ = drift or expected return (optional, typically set to 0)

- Δt = time increment

Liquidity Decay using Beta Distribution:

$$StressedDepth_i = InitialDepth * (1 - \lambda_i)$$

$$\lambda_i \sim Beta(\alpha, \beta)$$

a) Probability Density Function of the beta distribution:

$$f(x; \alpha, \beta) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)} \quad \text{for } x \in (0, 1)$$

b) Beta Function (expressed in terms of the Gamma Function):

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

where:

- α = how strongly the distribution is “pulled” to 1
- β = how strongly the distribution is “pulled” to the right

Collateral Shortfall:

$$LTV_i = \frac{Debt_i}{CollateralAmount_i * P_t}$$

$$Slippage_i = \eta * \sqrt{\frac{TotalLiquidationVolume}{StressedDepth_i}}$$

$$RecoveredDebt_i = LiquidatedCollateralAmount_i * P_t (1 - Slippage_i - LiquidationDiscountPenalty)$$

$$CollateralShortfall_i = \max(0, Debt_i - RecoveredDebt_i)$$

where:

- P_t = Price of asset at time t

OperationalScore ($a^{\rightarrow}$):

Let:

- $a^{\rightarrow} = [a_1, a_2, \dots, a_n]$ be the vector of proposed capital allocations across n markets
- $c^{\rightarrow} = [c_1, c_2, \dots, c_n]$ be the vector of ScaledCapitalCeilings (SCC) across n markets

Constrained by:

- $0 \leq a_i \leq c_i$ for all i and $\sum_i a_i \leq TotalCapital$

For vault i with the full allocation vector $a^{\rightarrow}$

$$OperationalScore(a^{\rightarrow})_i = w_1 S_{alloc}(a^{\rightarrow})^i + w_2 S_{util}(a^{\rightarrow})^i + w_3 S_{rate}(a^{\rightarrow})^i + w_4 S_{buffer}(a^{\rightarrow})$$

$$w_1 + w_2 + w_3 + w_4 = 1$$

$$a) \quad S_{alloc}(a^{\rightarrow}) = \frac{1}{n} \sum_{i=1}^n \left\{ 1 - \frac{|(a^{\rightarrow})_i - SCC_i^{\rightarrow}|}{\Delta A_{max}} \right\} \quad \text{if } a_i^{\rightarrow} \leq SCC_i$$

$$S_{alloc}(a^{\rightarrow}) = \frac{1}{n} \sum_{i=1}^n \{ 0 \} \quad \text{if } a_i^{\rightarrow} > SCC_i$$

where:

- $a_i^{\rightarrow}$ is the proposed allocation vector to vault i
- $SCC_i^{\rightarrow}$ is the scaled capital ceiling vector for vault i
- ΔA_{max} is a normalization constant - set to SCC_i

$$b) \quad S_{util}(a^{\rightarrow}) = \frac{1}{n} \sum_{i=1}^n \left[1 - \frac{|\hat{U}_i(a^{\rightarrow})_i - U_i^*|}{\Delta U_{max}} \right]$$

where:

- $\hat{U}_i(a^{\rightarrow})_i$ is the projected utilization for vault i under allocation $a^{\rightarrow}$
- U_i^* is the target utilization
- ΔU_{max} is the max allowed deviation - used for normalization
- $0 \leq S_{util}(a^{\rightarrow}) \leq 1$

$$c) \quad S_{rate}(a^{\rightarrow}) = \frac{1}{n} \sum_{i=1}^n \left[1 - \frac{|\hat{R}_i(a^{\rightarrow})_i - R_i^*|}{\Delta R_{max}} \right]$$

where:

- $\hat{R}_i(a^{\rightarrow})_i$ is the projected borrow rate for vault i under allocation $a^{\rightarrow}$
- R_i^* is the target borrow rate
- ΔR_{max} is the max allowed deviation - used for normalization
- $0 \leq S_{rate}(a^{\rightarrow}) \leq 1$

$$d) S_{buffer}(a^{\rightarrow}) = \frac{1}{n} \sum_{i=1}^n \left[1 - \frac{|\dot{B}_i(a^{\rightarrow}_i) - B_i^*|}{\Delta B_{max}} \right]$$

where:

- $\dot{B}_i(a^{\rightarrow}_i)$ is the projected borrow rate for vault i under allocation $a^{\rightarrow}$
- B_i^* is the target borrow rate
- ΔB_{max} is the max allowed deviation - used for normalization
- $0 \leq S_{buffer}(a^{\rightarrow}) \leq 1$

Risk Metric Normalization:

Let:

- p_{ij} , $i \in [M]$, $j \in [N]$ denote the initial value of the j -th metric of the i -th asset
- b_{ij} is the normalized value of p_{ij}
- $p_{min,j}$, $p_{max,j}$ are the minimum and maximum values of the j -th metric over all assets.

For risk metrics that increase with improving asset quality, normalization is as follows:

$$b_{ij} = \frac{p_{ij} - p_{min,j}}{p_{max,j} - p_{min,j}} * 100, \quad i \in [M], j \in [N]$$

For risk metrics that decrease with improving asset quality, normalization is as follows:

$$b_{ij} = \frac{p_{max,j} - p_{ij}}{p_{max,j} - p_{min,j}} * 100, \quad i \in [M], j \in [N]$$

Where $p_{min,j}$ = 2nd percentile, $p_{max,j}$ = 98th percentile

Area Under the Receiver Operating Characteristic Curve:

$$AUC = \frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n 1(s_i > s_j)$$

where:

- m : Number of positive (risky) examples
- n : Number of negative (safe) examples
- s_i : CRS score of positive example i
- s_j : CRS score of negative example j
- $1(s_i > s_j)$: Indicator function (1 if true, 0 otherwise)

Mean Separation:

$$MeanSeparation = |\mu_{risky} - \mu_{safe}|$$

where:

- μ_{risky} = Mean CRS score for risky markets
- μ_{safe} = Mean CRS score for safe markets

Precision @ Top K

$$Precision@TopK = \frac{1}{K} \sum_{i=1}^K y_i$$

where:

- K : Number of top-ranked markets
- $y_i \in \{0,1\}$: Ground truth label (1 = risky, 0 = safe) for the i th ranked market
- Markets are ranked by descending CRS score

References:

- [1] Rockafellar, R. T., & Uryasev, S. (2000). Optimization of Conditional Value-at-Risk. *Journal of Risk*, 2(3)
- [2] Amihud, Y. (2002). Illiquidity and stock returns: cross-section and time-series effects. *Journal of Financial Markets*, 5(1), 31–56. [https://doi.org/10.1016/S1386-4181\(01\)00024-6](https://doi.org/10.1016/S1386-4181(01)00024-6)
- [3] Charles, V., Gherman, T., & Paliza, J. C. (2022). The Gini Index: A Modern Measure of Inequality. In V. Charles, & A. Emrouznejad (Eds.), *Modern Indices for International Economic Diplomacy*, 55-84. Palgrave Macmillan. https://doi.org/10.1007/978-3-030-84535-3_3
- [4] Bradley, A. P. (1997). The use of the area under the ROC curve in the evaluation of machine learning algorithms. *Pattern Recognition*, 30(7), 1145–1159. [https://doi.org/10.1016/S0031-3203\(96\)00142-2](https://doi.org/10.1016/S0031-3203(96)00142-2)
- [5] Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to Information Retrieval* (p. 155, Section 8.3). Cambridge University Press. <https://nlp.stanford.edu/IR-book/>
- [6] Fisher, R. A. (1936). *The use of multiple measurements in taxonomic problems*. *Annals of Eugenics*, 7(2), 179–188
- [7] Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A Next-generation Hyperparameter Optimization Framework. *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2623–2631. <https://arxiv.org/abs/1907.10902>
- [8] Glasserman, P. (2003). *Monte Carlo Methods in Financial Engineering*. 481-535. Springer.
- [9] Glasserman, P. (2003). *Monte Carlo Methods in Financial Engineering*. 93-104. Springer.
- [10] Cunningham, A. (n.d.). *The Beta Distribution*. Retrieved from <https://www.acsu.buffalo.edu/~adamcunn/probability/beta.html>
- [11] Gauntlet. (2024). *Vault Curation Considerations: A Deeper Dive – Asset Liquidity*. Retrieved from <https://vaultbook.gauntlet.xyz/morpho-vaults/vault-curation-considerations-a-deeper-dive/asset-liquidity>
- [12] Mars Protocol. (2023). *Protocol Risk Framework*. Retrieved from <https://docs.marsprotocol.io/risk-methodology/mars-risk-framework>